

Project Workflow: Week 3

Phase: Environment Setup & Project Initialization

1. Objective

The goal of Week 3 was to prepare the development environment and initialize the project structure so that implementation could begin. This included:

1. Installing and configuring the JDK and IDE for Java development.
2. Configuring the Java Swing/AWT environment.
3. Creating the project structure and the CarRacingGame JFrame entry point.

2. Detailed Task Breakdown

▪ Installing the JDK and Configuring the IDE

Installed the Java Development Kit (JDK 8+) and set up the chosen IDE (IntelliJ IDEA / VS Code) for development. This involved verifying the JDK installation and confirming that a simple test program could be compiled and run successfully.

▪ Configuring the Java Swing/AWT Environment

Set up the Swing and AWT libraries needed for the game's window and input handling. This ensured the project was ready to support a JFrame-based window and a javax.swing.Timer-based game loop, all of which are required for later weeks of implementation.

▪ Creating the Project Structure and Entry Point

Created the initial project folder structure to organise source files as the codebase grows, and implemented the CarRacingGame class as a JFrame. This class serves as the main entry point of the application, responsible for creating and displaying the game window that will later hold the GamePanel and the rest of the game components designed in Week 2.

3. Deliverables

1. Configured Development Environment: JDK installed and IDE set up and verified for Java development.
2. Swing/AWT Setup: Project environment configured to support Swing/AWT and input.

3. Initial Project Structure: Base folder/package structure created for the codebase.
4. CarRacingGame Entry Point: A working JFrame class that launches and displays the game window.

4. Tools & Resources

- **Development Environment**

1. Java JDK 8+
2. IntelliJ IDEA / VS Code

- **Reference Materials**

Java Swing and AWT setup guides, JDK installation documentation, and IDE configuration references for Java desktop application development.